[Prop. Normal and Nilpotent $\Rightarrow$ Zero.
Proof. Let $T: V \to V$ be a linear operator on fin. dim inn. prod space V.
Since T is normal, there exists a basis $\beta = \{\alpha_1, \cdots, \alpha_n\}$ for V consisted by eigenvectors of T. Given $\alpha \in V$,

$$\alpha = x_1\alpha_1 + \cdots + x_n\alpha_n \quad \text{for some } x_i \in F$$

and so $T\alpha = x_1c_1\alpha_1 + \cdots + x_nc_n\alpha_n$, where c_i are eigenvalues corresponding to α_i.
Meanwhile, since T is nilpotent, all eigenvalues for T are zero, so $T\alpha = 0$.
Thus, $T \equiv 0$.

Alternatively, without spectral decomposition, observe that: if T is normal,

$$\|T\alpha\|^2 = \langle T\alpha, T\alpha\rangle = \langle \alpha, T^*T\alpha\rangle = \langle \alpha, TT^*\alpha\rangle = \langle T^*\alpha, T^*\alpha\rangle = \|T^*\alpha\|^2$$

so $T\alpha = 0 \Leftrightarrow T^*\alpha = 0$, which gives $\ker T = \ker T^*$.
Now, suppose that T is nilpotent and $T \neq 0$. Take $m \in \mathbb{N}$ be the smallest integer such that $T^m \equiv 0$. (That is, $T^0, \cdots, T^{m-1} \neq 0$).
Let $\alpha \in V$ such that $T^{m-1}\alpha \neq 0$. Then, since $T^m = 0$, $T^{m-1}\alpha \in \ker T$.
Since $\ker T = \ker T^*$, $T^*T^{m-1}\alpha = 0$. So,

$\langle T^*T^{m-1}\alpha,$

[Prop.
Let T be a normal operator on a fin. dim complex inn. prod space V.
Then, T^* is polynomial in T.
Proof. Since T is normal, there is a spectrum decomposition

$$T = c_1 T_1 + \cdots + c_k T_k$$

where c_i are disjoint eigenvalues of T and
T_i are orthogonal projection onto $\ker(T - c_i I)$.
So, $T^* = \overline{c_1} T_1^* + \cdots + \overline{c_k} T_k^* = \overline{c_1} T_1 + \cdots + \overline{c_k} T_k$,

Meanwhile, since each T_i's are orthogonal, given polynomial $f(t) \in \mathbb{C}[t]$,

$$f(T) = f(c_1) T_1 + \cdots + f(c_k) T_k,$$

therefore let

$$f = \sum_{i=1}^{k} \overline{c_i} \cdot \prod_{j \neq i} \frac{(x - c_j)}{(c_i - c_j)}$$

Then, f is $\mathbb{C}$-polynomial and

$$f(c_i) = \overline{c_i}$$

So that

$$f(T) = T^*, \text{ as we desired.} \quad \square$$